

Broken Heart Syndrome

Dying of a broken heart is a tale as old as civilisation but scientists have now uncovered the facts behind the lore

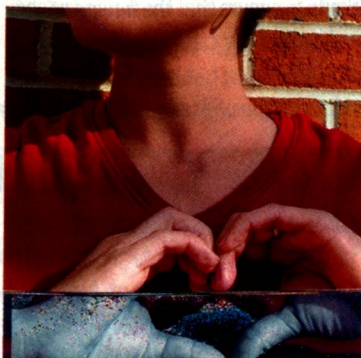


New Research

A 2010 article in *European Heart Journal* identifies for the first time the physio-pathological mechanism of the disease. The researchers contend that while BHS is not as common as a typical heart attack, it possibly occurs more frequently than doctors realise. They expect the numbers to increase as more physicians learn to recognise the syndrome's unique features.

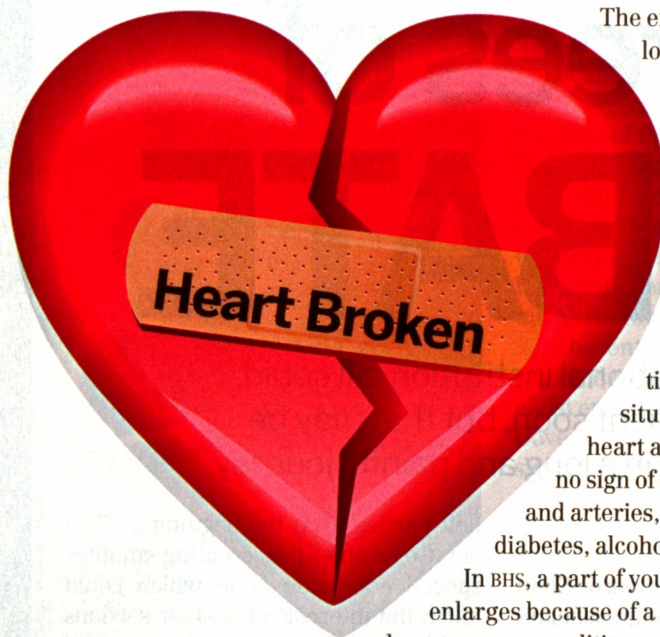
99%

OF THOSE WHO GET
BROKEN HEART
SYNDROME ARE
LIKELY TO SURVIVE.



WOMEN'S WOE

BHS affects mostly women in the post-menopause period, when they are no longer protected by their estrogen hormones.



The end of a relationship, the loss of a job, the death of a loved one—all can break your heart. Really. Scientists now have an ear-catching name for a heart in the grip of emotional distress: Broken Heart Syndrome (BHS). It's a temporary heart condition brought on by stressful situations which mimics a heart attack but with little or no sign of deterioration of the heart and arteries, cholesterol build-up, diabetes, alcoholism, embolism etc.

In BHS, a part of your heart temporarily enlarges because of a surge in stress hormones—a condition called cardiomyopathy.

MIMICKING A HEART ATTACK

Sudden emotional stress can result in symptoms that mimic a classic heart attack



Acute pain in the chest and shortness of breath exactly like in a heart attack.

ECG also shows clinical changes seen typically in a heart attack.

But ECG also reveals that the tip of the heart chamber bulges in BHS, unlike in a heart attack.

Angiogram and MRI scans show no blockages and irreversible muscle damage.

A STUNNED HEART

In BHS, stress hormones temporarily 'stun' the heart. Researchers say that some people respond to sudden, overwhelming emotional stress by releasing large amounts of catecholamines (epinephrine and norepinephrine) into the bloodstream, along with their breakdown products and small proteins produced by an excited nervous system. These chemicals can be temporarily toxic to the heart, effectively stunning the muscle and producing symptoms similar to a typical heart attack.



■ Compiled by **Damayanti Datta**